DM I: Block 'Model Evaluation and Model Comparisons'

Unit 2: Finding the best model

Myra Spiliopoulou



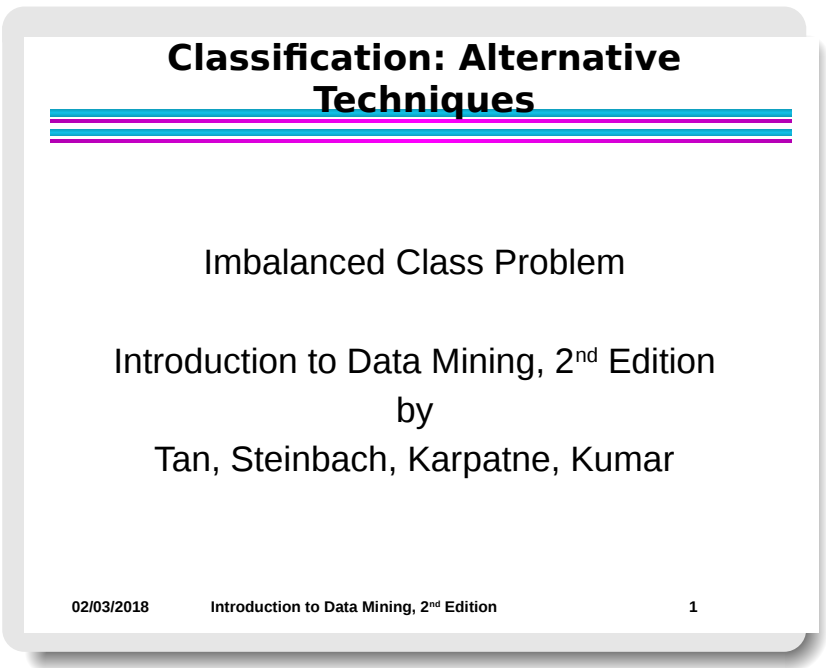


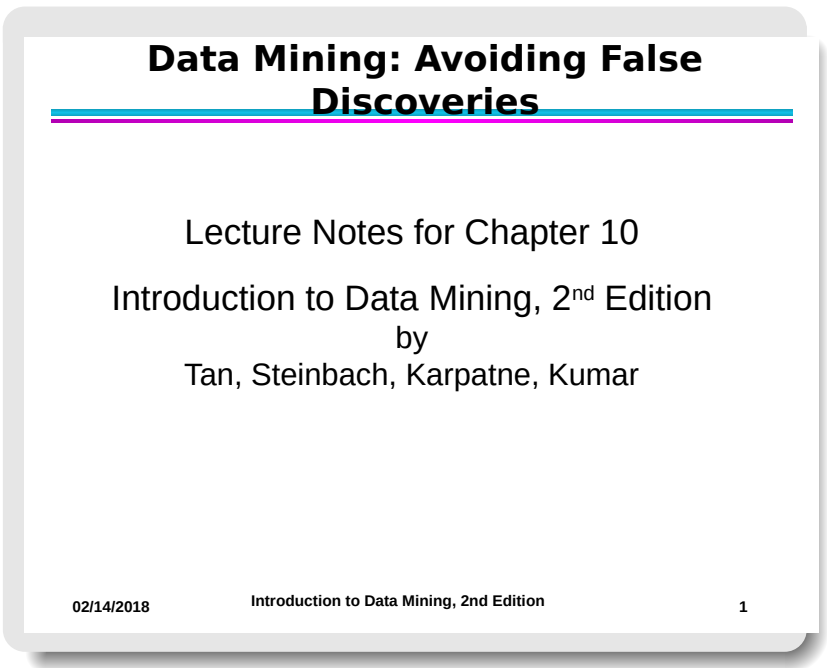
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Sources of the unit

- Book of the course (edition 2019), chapter 10
- Old edition of the course, chapter 4, subchapter 4.6
- Gama, J., Zliobaite, I., Bifet, A., Pechenizkiy, M., and Bouchachia, A. (2014). *A survey on concept drift adaptation.* ACM Comput. Surv., 46(4)
- Slideset of **2018**, chapter 4
- Slideset of **2018**, chapter 10 – font and layout modified in several slides; correction on 1st item of slide 11





1 Good and Best

2 Comparing two models

3 Role of the test sample size

4 Comparing many models

5 Flashback: cost-awareness

6 Closing

RECALL: Evaluation Unit 1

We have seen:

- $\surd$ different ways of modeling classifier quality
- $\surd$ methods for building training & test samples by sampling (a) without replacement and (b) with replacement
- $\surd$ ways of combining classifiers to improve classification quality
- $\surd$ ways of dealing with class imbalance

We have seen many ways of building a single/ensemble classifier:

- How to figure out which classifier is the best one?

Good model, best model

Best model

Given are n models $M_1, \ldots, M_n$ trained on D_{train} and tested on D_{test}. Which is the best one among them?

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Given is a model M trained on D_{train} and tested on D_{test}. How to find whether M is good?

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Best model

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Good model

Given is a model M trained on D_{train} and tested on D_{test}. How to find whether M is good?

Define: 'good' $\equiv$ 'better than chance'

RECALL: Classifier M (Unit 1)

IF I2=m THEN Response=no
ELSEIF I26=no THEN Response=yes ELSE Response=no

D_{test}

Id	I2	I15	I30	I22	I31	I9	I26	Response
#91	f	VH	yes	better	no	r	no	yes
#92	m	M	no	better	no	b	yes	yes
#93	m	M	no	worse	no	b	yes	no
#94	f	VH	yes	worse	no	b	yes	yes
#95	m	L	no	no effect	no	l	no	no
#96	m	M	no	better	no	l	no	no
#97	f	VH	yes	better	yes	l	no	yes
#98	f	H	no	better	yes	r	yes	no
#99	f	H	yes	better	no	l	no	yes

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#96	m	M	no	better	no	l	no	no
#97	f	VH	yes	better	yes	l	no	yes
#98	f	H	no	better	yes	r	yes	no
#99	f	H	yes	better	no	l	no	yes

M assigns:

$\widehat{\text{Response}}$
yes
no
no
no
no
no
yes
no
yes

- M: accuracy = $\frac{7}{9}$ F-measure = $\frac{2\cdot\frac{3}{3+0}\cdot\frac{3}{5}}{\frac{3}{3+0}+\frac{3}{5}} = \frac{6/5}{8/5} = \frac{6}{8} = 0.75$
- A model M_0 that assigns each instance to the majority class: ?
- A model M_1 that classifies by tossing a biased coin: class 'no' comes with likelihood $\frac{10}{15}$: ?

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Evaluating a model on a sample

- **Is it good?** $\Rightarrow$ Comparison with a baseline model of known performance – e.g. with the random classifier [1]
- **Is it better?** $\Rightarrow$ Comparison with another model
  Option 1: both learned on the same sample
  Option 2: learned on different samples

[1] WARNING: Sometimes, there are more than one ways of defining **a** random classifier.

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Relying on the difference between two numbers

When we compare two models with some performance function, the performance values look different. But are they truly different?

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$\Downarrow$

Statistical testing

NULL hypotheses & tests

Slideset from Ch. 10 on "statistical testing"

SLIDES 10-15 (basics and an example test), 16-17 (one-sided and two-sided tests)

[1] WARNING: Sometimes, there are more than one ways of defining **a** random classifier.

Statistical Testing …

- Make inferences (decisions) about that validity of a result
- For statistical inference (testing), we need two thing:
  - A statement that we want to disprove
    - Called the **null hypothesis (H_0)**
    - The null hypothesis is typically a statement that the result is merely due to random variation
    - It is typically the opposite of what we would like to show
  - A random variable, , called a **test statistic**, for which we know or can determine a distribution if H_0 is true.
    - The distribution of under H0 is called the **null distribution**
    - The value of is obtained from the result and is typically numeric

Examples of Null Hypotheses

- A coin or a dice is a fair coin.
- The difference between the means of two samples is 0
- The purchase of a particular item in a store is unrelated to the purchase of a second item, e.g., the purchase of bread and milk are unconnected
- The accuracy of a classifier is no better than random

Significance Testing

- Significance testing was devised by the statistician Fisher
- Only interested in whether null hypothesis is true
- Significance testing was intended only for exploratory analyses of the null hypothesis in the preliminary stages of a study
  - For example, to refine the null hypothesis or modify future experiments
- For many years, significance testing has been a key approach for justifying the validity of scientific results

How Significance Testing Works

- Analyze the data to obtain a result
  - For example, data could be from flipping a coin 10 times to test its fairness
- The result is expressed as a value of the test statistic, R
  - For example, let R be the number of heads in 10 flips
- Compute the probability of seeing the current value of R or something more extreme
  - This probability is known as the **p-value** of the test statistic

How Significance Testing Works …

- If the p-value is sufficiently small, we reject the null hypothesis, H_0 and say that the result is statistically significant
  - We say we reject the null hypothesis, H_0
  - A threshold on the p-value is called the **significance level, α**
    - Often the significance level is 0.01 or 0.05
- If the p-value is **not** sufficiently small, we say that we fail to reject the null hypothesis
  - Sometimes we say that we accept the null hypothesis but a high p-value does not necessarily imply the null hypothesis is true

Example: Testing a coin for fairness

- H_0: $P(X=1) = P(X=0) = 0.5$
- Define the test statistic R to be the number of heads in 10 flips
- Set the significance level α to be 0.05
- The number of heads R has a binomial distribution
- For which values of R would you reject H_0?

k	$P(S = k)$
0	0.001
1	0.01
2	0.044
3	0.117
4	0.205
5	0.246
6	0.205
7	0.117
8	0.044
9	0.01
10	0.001

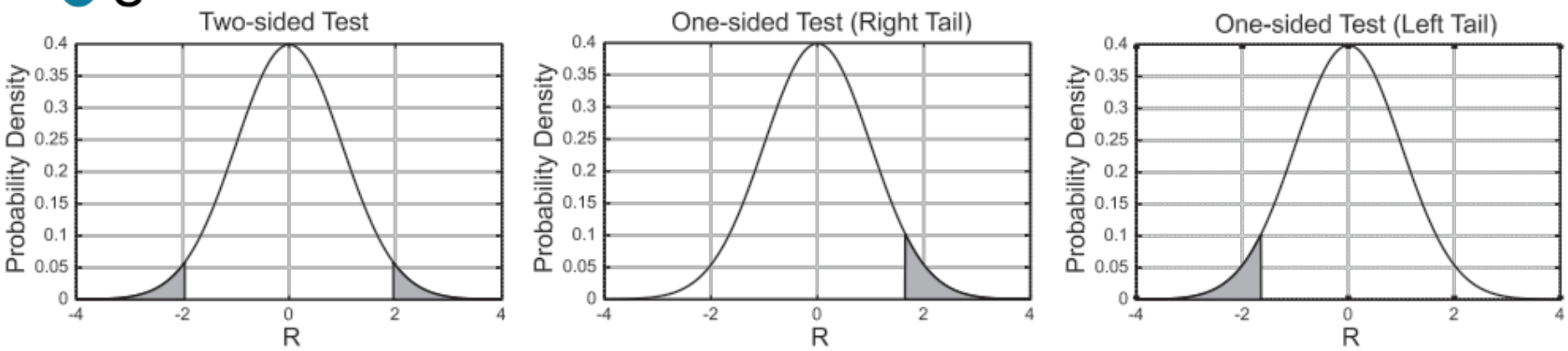
One-sided and Two-sided Tests

- More extreme can be interpreted in different ways
- For example, an observed value of the test statistic, R_{obs}, can be considered extreme if
  - it is greater than or equal to a certain value, R_H ,
  - smaller than or equal to a certain value, R_L, or
  - outside a specified interval, $[R_H, R_L]$.
- The first two cases are “one-sided tests” (right-tailed and left-tailed, respectively),
- The last case results in a “two-sided test.”

One-sided and Two-sided Tests …

- Example of one-tailed and two tailed tests for a test statistic R that is normally distributed for a roughly 5% significance level.

- S



Statistical testing for the evaluation of a model M

$q()$: quality function
M_1: 'baseline' or 'competitor' classifier on the same data as M
H_0: on the difference between $q(M)$ and $q(M_1)$. [The R]

Statistical testing for the evaluation of a model M

$q()$: quality function
M_1: 'baseline' or 'competitor' classifier on the same data as M
H_0: on the difference between $q(M)$ and $q(M_1)$. [The R]

When $q()$ is on the number of misclassified instances

McNemar test from Section 4.2.3 of (Gama et al, 2014) [a]

- **H_0:** Classifiers M, M_1 perform equally
- **to be rejected** with confidence δ depending on the value of

$$mcn = \frac{sign(a-b) \times (a-b)^2}{a+b}$$ where

a: number of instances misclassified by M but not by M_1
b: number of instances misclassified by M_1 but not by M

"The test follows the χ^2 distribution. At 0.99 confidence it rejects the null hypothesis (the performances are equal) if $mcn > 6.635$."

[a] with clarifications and modifications

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Models on test samples of different sizes S. 4.6 (old edition)

Example

Given are two classifiers, M_A, M_B:

- M_A has been tested on 30 records. The accuracy is 0.85.
- M_B has been tested on 5000 records. The accuracy is 0.75.

Which classifier is better?

Models on test samples of different sizes S. 4.6 (old edition)

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Comparing two classifiers in a reliable way

- How much *confidence* do we place in each classifier's performance?
- To what extend can we explain the differences in performance between/among the classifiers through variations in the composition of the test sample?

Exploiting classifier confidence in comparisons 4.6.1

Classifier confidence:

Binomial experiment consisting of n independent trials

- One (Bernoulli) trial has two possible outcomes: success (guessing the label correctly) & failure; it follows the binomial distribution with mean number of successes p and variance $p(1-p)$.
- Let X be the number of successes in n trials. Then, for $m \leq n$, $P(X=m)$ follows the binomial distribution with mean np and variance $np(1-p)$:

$$P(X=m) = \left(\begin{array}{c} n \\ m \end{array} \right) p^m (1-p)^{n-m}$$

- If n is sufficiently large, the normal distribution can be used instead.

Classifier confidence

4.6.1, Witten & Eibe

Classifier confidence, assuming normal distribution:

Confidence interval for given confidence level — S. 4.6 (old edition) (Eqs. 4.12, 4.13)

The accuracy acc at a confidence level $1-a$ is (cf. Eq. 4.12):

$$P(-Z_{a/2} \leq \frac{acc-p}{\sqrt{\frac{p(1-p)}{n}}} \leq Z_{1-a/2}) = 1-a$$

where $Z_{a/2}, Z_{1-a/2}$ are the upper and lower bounds obtained from a standard normal distribution at confidence level $1-a$.

Confidence interval for given confidence level — Witten & Eibe

Given is a random variable X with mean=0 and variance=1. $P(-z \leq X \leq z)$ is the confidence $c = 1-a$ with which $X \in [-z, z]$.
For a symmetric distribution, it holds that $c = 1-2P(X > z)$, i.e. $\frac{a}{2} = P(X > z)$.

To perform the computations for the accuracy acc, we must derive a variable X from acc that has the required mean and variance.

Classifier confidence under normal distribution Witten & Eibe

Under normal distribution, the z-values for various confidence levels are:

Confidence c	99%	98%	90%	80%	60%
$a/2$	0.005	0.01	0.05	0.1	0.2
z	2.58	2.33	1.65	1.28	0.84

Example, partially from Witten & Eibe

We perform n trials and observe 75% successes, i.e. the accuracy is 75%.
What is the confidence interval of accuracy?

trials n	1000
successes	750
acc is in	[73.2,76.7]

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trials n	100
successes	75
acc is in	[69.1,80.1]

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acc is in	[73.2,76.7]

trials n	100
successes	75
acc is in	[69.1,80.1]

trials n	2000
successes	1500
acc is in	[??, ??]

trials n	40
successes	30
acc is in	[??, ??]

trials n	20
successes	15
acc is in	[??, ??]

Comparing the error rates of two models on independent test sets 4.6.2

Let M_A, M_B be two models:

M_A: error rate e_A (observed number of failures) on test set D_A ($|D_A| = n_A$)

M_B: error rate e_B (observed number of failures) on test set D_B ($|D_B| = n_B$)

where n_A, n_B are sufficiently large to assume normal distribution of the error rates.
Let $d = e_A - e_B$

- d is also normanlly distributed, with mean d_t and variance σ_d^2.
- The variance σ_d^2 can be approximated by the sum of the variances of the error rates:

$$\widehat{\sigma}_d^2 = \frac{e_A(1-e_A)}{n_A} + \frac{e_B(1-e_B)}{n_B}$$

- At confidence level $c = 1 - a$, the confidence interval for the true difference d_t is $[d - z_{a/2}\widehat{\sigma}_d, d + z_{a/2}\widehat{\sigma}_d]$.

Comparing the error rates of two models on independent test sets 4.6.2

Recall the example

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Comparing the error rates of M_A, M_B 4.6.2

The null hypothesis says that $d_t = 0$.

M_A: error rate $e_A =$ on a test set of size $n_A = 30$.

M_B: error rate $e_B =$ on a test set of size $n_B = 5000$.

- $d = |e_A - e_B| =$ and $\widehat{\sigma}_d^2 =$
- For $c = 95\%$, $\zeta = 1.96$, so that $d_t =$

If the interval spans the value ZERO, we conclude that the d_t is not significant at confidence level c.

Comparing the error rates of two models on independent test sets 4.6.2

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Confidence level at which the null hypothesis must be rejected 4.6.2

- Compute z
- For this z, compute c (highest possible for two-sided test).

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Comparing many algorithms on many datasets (Gama et al, 2014)

Comparing $n \geq 2$ algorithms on m datasets:

1. **Friedman test:** on **H_0**: "the learners perform equally"
2. If and only if H_0 is rejected by the Friedman test, then Nemenyi test on **H_0**: "learners M_A, M_B perform equally", where M_A, M_B are drawn from the n classifiers.

"The Nemenyi test is the following: two classifiers are performing differently if the corresponding average ranks [over all m datasets] differ by at least the critical difference $CD = q_\alpha \sqrt{n(n+1)/6m}$ where n is the number of learners, m is the number of datasets, and critical values q_α are based on the Studentized range statistic divided by $\sqrt{2}$."

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RECALL: Class imbalance and class misattribution cost

- One class is much more rare than the other(s).
- The cost of a wrong class attribution varies substantially among the classes.

Often there is one class that satisfies both conditions.



Learning under class imbalance

- Oversample from the rare class
- Undersample from the non-rare class(es)
- Do both
- Induce rules that focus on the rare class

Evaluating under class imbalance

F-measure instead of accuracy

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F-measure instead of accuracy

Taking class misattribution cost into account

Enhance the confusion matrix with weights

Slideset from Ch. 4 on “Imbalanced classes”

SLIDES 25 (ways of dealing with imbalanced classes), 26 (cost matrix), 27 (example), 28-29 (cost-sensitive classification)

Handling Class Imbalanced Problem

- Class-based ordering (e.g. RIPPER)
  - Rules for rare class have higher priority
- Cost-sensitive classification
  - Misclassifying rare class as majority class is more expensive than misclassifying majority as rare class
- Sampling-based approaches

Cost Matrix

	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	f(Yes, Yes)	f(Yes,No)
	Class=No	f(No, Yes)	f(No, No)

Cost Matrix	PREDICTED CLASS		
ACTUAL CLASS	C(i, j)	Class=Yes	Class=No
	Class=Yes	C(Yes, Yes)	C(Yes, No)
	Class=No	C(No, Yes)	C(No, No)

C(i,j): Cost of misclassifying class i example as class j

$$\text{Cost} = \sum C(i, j) \times f(i, j)$$

Computing Cost of Classification

Cost Matrix	PREDICTED CLASS		
	C(i,j)	+	-
ACTUAL CLASS	+	-1	100
	-	1	0

Model M_1	PREDICTED CLASS		
		+	-
ACTUAL CLASS	+	150	40
	-	60	250

Model M_2	PREDICTED CLASS		
		+	-
ACTUAL CLASS	+	250	45
	-	5	200

Accuracy = 80%

Cost = 3910

Accuracy = 90%

Cost = 4255

Cost Sensitive Classification

- Example: Bayesian classifer
  - Given a test record x:
    - Compute p(i|x) for each class *i*
    - Decision rule: classify node as class k if

$$k = \arg\max_i p(i \mid x)$$

  - For 2-class, classify x as + if p(+|x) > p(-|x)
    - This decision rule implicitly assumes that C(+|+) = C(-|-) = 0 and C(+|-) = C(-|+)

Cost Sensitive Classification

- General decision rule:
  - Classify test record x as class k if

$$k = \arg\min_j \sum_i p(i \mid x) \times C(i, j)$$

- 2-class:
  - Cost(+) = p(+|x) C(+,+) + p(-|x) C(-,+)
  - Cost(-) = p(+|x) C(+,-) + p(-|x) C(-,-)
  - Decision rule: classify x as + if Cost(+) < Cost(-)
    - if C(+,+) = C(-,-) = 0:

$$p(+ \mid x) > \frac{C(-,+)}{C(-,+) + C(+,-)}$$

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Summary and Outlook

In these two units we have seen:

- √ How to quantify the performance of a classifier
- √ How to compute the performance value for a classifier, given a set D to be used for training and testing
- √ How to deal with classes that are more rare than others, and/or
- √ have different costs of class misattribution
- √ How to compare two classifiers, and
- √ what is the role of the size of the test set
- √ How to compare classification algorithms on different datasets

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- √ How to compare classification algorithms on different datasets

Next:

We move from supervised to unsupervised learning and discuss clustering algorithms – a 'model' is then a 'set of clusters over all the data'.

Thank you very much!

Questions?